

Prateek Kumar Sahu

PhD Candidate

Department of Psychology

University of Alberta, CA

Proficiency in coding languages: **R** (Expert), Python (Moderate)

Proficient in Causal inference, **Bayesian** modeling, **Bio-acoustics**

Contact

psahu@ualberta.ca

<https://prateeksahu.com>

[Linkedin](#), [GitHub](#)

Education

- University of Alberta** Edmonton, Canada
PhD, Department of Psychology 2022 - Now
- University of Alberta** Edmonton, Canada
MSc, Department of Psychology 2019 - 2022
- Indian Institute of Science Education and Research** Kolkata, India
BSc & MSc, Department of Biological Sciences 2013 - 2018

Research Experience

- Animal cognition, Vocal production and perception, Novel bioacoustic techniques, open-source research hardware design, preferences in songbirds *Dr.Christopher Sturdy* Doctoral project 2022-now
- Methodological comparisons of classification techniques and refinement of an operant go/no-go procedure *Dr.Christopher Sturdy* Masters' thesis 2019-2022
- Neural basis of sugar elicited search behaviour of Drosophila *Dr.Axel Brockmann* Research Project 2018-2019
- Understanding reunification dynamics in ant colonies *Dr.Sumana Annagiri* Summer Project, Masters' thesis 2015,2017-2018
- Consistency in personality traits of the shoals of climbing perch *Dr.V. V. Binoy* Summer Project 2016

Publications

10. **Sahu, P.K.**, Deimeke, M.J., Garcia, A.D., Stenstrom, K.H., Smeltz, S.M.L., Wang, Y., Ratch, J., & Sturdy, C.B. (2026) Re-examining hidden fitness: Female preferences for long-path songs in zebra finches. PLOS ONE, 21(3):e0343886, <https://doi.org/10.1371/journal.pone.0343886>
9. **Sahu, P.K.**, Garcia, A.D., Stenstrom, K.H., Deimeke, M.J., Smeltz, S.M.L., Lank, I.J., & Sturdy, C.B. (2025) A cost-effective Raspberry pi-based operant playback setup to evaluate auditory preferences in songbirds. Behavioural Processes, 232, 105276, <https://doi.org/10.1016/j.beproc.2025.105276>

8. Smeltz, S.M.L., Deimeke, M.J., Montenegro, C., **Sahu, P.K.**, Stenstrom, K.H., Camacho-Alpizar, A., & Sturdy, C.B. (2025) Individual discrimination within, but not between, two vocalization types of the black-capped chickadee. *Ethology*, e13572. <https://doi.org/10.1111/eth.13572>
7. Smeltz, S.M.L., Deimeke, M.J., **Sahu, P.K.**, Montenegro, C., Stenstrom, K.H., Starenchak, I., Rennie, V., Ebrahim, I.M., Jones, J.A., & Sturdy, C.B. (2025) Individual vocal identity may be obscured following colony assembly in captive black-capped chickadees. *Behavioural Processes*, 225, 105151, <https://doi.org/10.1016/j.beproc.2025.105151>
6. **Sahu, P.K.**, Montenegro, C., Lambert, C.T., Oprea, A., Deimeke, M.J., Rennie, V., Smeltz, S.M.L., Benowicz, T.J., Patel, D., Phillmore, L.S., & Sturdy, C.B. (2022) Effect of feed-time duration on discrimination of vocalizations in a go/no-go operant paradigm. *Behavioural Processes*, 203, 10477, <https://doi.org/10.1016/j.beproc.2022.104777>
5. **Sahu, P.K.**, Campbell, K.A., Oprea, A., Phillmore, L.S., & Sturdy, C.B. (2022) Comparing methodologies for classification of zebra finch distance calls. *The Journal of the Acoustical Society of America*, 151, 3305, <https://doi.org/10.1121/10.0011401>
4. Lambert, C.T., **Sahu, P.K.**, Sturdy, C.B., & Guillette, L.M. (2022) Among-individual differences in auditory and physical cognitive abilities in zebra finches. *Learning & Behaviour*, 50(3), 389-404, <https://doi.org/10.3758/s13420-022-00520-w>
3. Montenegro, C., **Sahu, P.K.**, & Sturdy, C.B. (2021). Individual differences in female black-capped chickadee (*Poecile atricapillus*) fee-bee songs. *The Journal of the Acoustical Society of America*, 150(4), 3038-3046, <https://doi.org/10.1121/10.0006532>
2. Montenegro, C., Service, W.D., Scully, E.N., Mischler, S.K., **Sahu, P.K.**, Benowicz, T.J., Fox, K.V. R., & Sturdy, C.B. (2021). The impact of anthropogenic noise on individual identification via female song in Black-capped chickadees (*Poecile atricapillus*). *Scientific Reports*, 11(1), <https://doi.org/10.1038/s41598-021-96504-3>
1. **Sahu, P.K.**, Kolay, S. & Annagiri, S. (2019) To reunite or not: A study of artificially fragmented *Diacamma indicum* ant colonies. *Behavioural Processes* 158, 4-10, <https://doi.org/10.1016/j.beproc.2018.10.017>

Teaching Experience & Service

- Teaching Assistant: Basic Psychological Processes (2019, 2020, 2021, 2022), Behaviour Modification (2020, 2021, 2022, 2023, 2024, 2025), Principle of Learning (2020), Behaviour & Genetics (2021), Neuroethology (2022), Honours Seminar: Writing course (2023), Theory in Learning and Comparative Cognition (2024), Social Influence (2024, 2025), Research Methods Cognition (2025), Human Neuropsychology (2025)
- Skype A Scientist: Talks at various schools (Indonesia, New York, Colorado, New York) (2022-2025)
- Secretary/Treasurer of the Graduate Psychology Association: (2023-2026)
- Judge for the Royce Research Conference, University of Alberta, CA: (2021, 2024, 2025)
- Volunteer for organizing the International Conference on Comparative Cognition, USA: (2023, 2025)

Awards & Scholarships

Alberta Graduate Excellence Scholarship (AGES), CA	2025-26
Graduate Student Travel Award, CA	2024-25
Comparative Cognition Society- Ron Weisman Travel Award, USA	2025
Alberta Graduate Excellence Scholarship (AGES), CA	2024-25
Alberta Graduate Excellence Scholarship (AGES), CA	2023-24
Scholarship from Department of Psychology, University of Alberta, CA	2019-22
INSPIRE Fellowship, IN	2013-18
National Means cum Merit Scholarship, IN	2008-10

Conferences

• Royce Research Conference	2025
• International Conference On Comparative Cognition	2025
• The Canadian Society for Brain, Behaviour and Cognitive Science (CSBBCS)	2024
• International Conference On Comparative Cognition	2023
• International Conference On Comparative Cognition	2021
• Animal Behaviour Society Conference	2020
• Virtual Royce Conference	2020
• Insect Multisensory Integration Workshop	2019
• 2nd Bangalore Meeting on Asian Bees	2019
• Preparatory School on Population Genetics and Evolution	2019
• Understanding Behaviour	2019
• Workshop on Cyberspace and Cyber Security	2017
• The Ambassadors of Evolution Workshop	2017
• The 10th Year Celebration of IISER Kolkata Advances in Life Sciences	2017
• The Frontiers in Modern Biology (FIMB) symposium	2015

Manuscript Reviews

1. *Ostrich* (2022); 2,3. *PeerJ* (2022, 2023), 4. *Behavioral Ecology and Sociobiology* (2022), 5. *Animal Cognition* (2023), 6. *Animal Behaviour* (2023), 7. *Animal Behaviour* (2023), 8. *The Journal of the Acoustical Society of America* (2023), 9. *Journal of Comparative Psychology* (2023), 10. *PLOS One* (2024), 11. *IBIS: International Journal of Avian Science* (2024), 12. *Journal of Experimental Biology* (2025), 13,14. *Behavioural Processes* (2025, 2025)

Skills

Bio-acoustical techniques & animal learning paradigms University of Alberta, 2019-now
Handling birds & animal care University of Alberta, 2019-now
Raising & handling *Drosophila* National Centre for Biological Sciences, 2018-19
Insect(ants, bees) research & insect biology IISER-K,NCBS 2015-19
Fish research & handling National Institute of Advanced Studies, 2016
Coding & Statistics "R", "Python", "Matlab"
Graphics Designing "Photoshop", "Illustrator"